

Youssef Alaa Alfahl

Full-Stack Developer | React, Node.js, Express, MongoDB, PostgreSQL | AI-Enabled Web Applications

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Professional Summary

Full-Stack Developer with hands-on experience building AI-enabled web and mobile applications using React, Vue.js, Next.js, Node.js, Express, MongoDB, PostgreSQL, ASP.NET Web API, FastAPI, and REST/WebSocket APIs. Built production-style projects covering authentication, media uploads, CRUD systems, social interactions, e-commerce workflows, semantic search, AI-powered summaries, and real-time client-backend communication. Strong background in software engineering, backend API development, database-integrated features, and AI-enhanced application development.

Experience

Ideal EG, Giza, Egypt *[Feb 2024 – Present]*

Sector: Software Development | **Role:** Full Stack Web Developer

- Developed full-stack web applications using React, Next.js, Node.js, Express.js, MongoDB, and PostgreSQL.
- Designed and consumed REST APIs implemented JWT authentication, file uploads, and role-based access control.
- Collaborated with cross-functional teams using Git/GitHub and participated in feature development, debugging, and deployment.
- Optimized database queries and application performance while maintaining clean, scalable code.

National University of Sciences and Technology (NUST), Islamabad, Pakistan *[Jul 2025 – Aug 2025]*

Sector: School of Electrical Engineering and Computer Science (SEECs) | **Role:** Research Intern

Project: Multitask Deep Learning for Real-Time Optimization of Polarization Snapshots in Rotating-Polarization BPSK/QPSK

- Simulation:** Modeled RP BPSK/QPSK links (AWGN/Rayleigh/Rician) and generated a Monte Carlo dataset in MATLAB.
- Modeling:** Labeled optimal N_p under latency/ops-per-bit constraints and trained a multitask DL model to select N_p and predict BER.
- Outcome:** Closely matched brute-force decisions with much lower computation, enabling real-time adaptive RP link configuration.

DHCB, Giza, Egypt *[Oct 2025 – Dec 2025]*

Sector: Software Development | **Role:** Full Stack Developer

Built and maintained frontend interfaces, backend APIs, and database-integrated features for software applications in a collaborative development environment, focusing on responsive UI, REST API integration, and full-stack feature delivery.

University of Science and Technology in Zewail City *[Apr 2026 – Jun 2026]*

Sector: Education | **Role:** Part-Time Teaching Assistant | **Course:** CSCI 101 – Introduction to Computer Science (Python)

Support undergraduate students by teaching Python programming and guiding them through core problem-solving concepts.

Education

B.S.C., Communications and Information Engineering, *University of Science and Technology in Zewail City*

Enrollment: 03/10/2021

Graduation: 01/02/2026

Final GPA: 3.80

Honors: MAGNA CUM LAUDE

Projects

Memento, Context-Aware Daily Task & Memory Assistant | Graduation Project

[Nov 2024 – Jun 2025]

- Built an AI-enabled assistant with a Flutter mobile app and FastAPI backend using WebSockets, PostgreSQL, Weaviate, LangGraph, and Gemini.
- Implemented user authentication, passive voice-based event capture, structured event extraction, semantic memory search, conflict detection, time/location reminders through notifications, and real-time client-backend communication.
- Designed backend AI workflows for routing, scheduling, updating, querying, summarization, and memory-aware responses using stateful agent orchestration.

EduBridge | Bilingual Online Educational Platform

- Built a full-stack online learning platform using Vue 3, Vite, Pinia, ASP.NET Core Web API, Entity Framework Core, and SQL Server, with a layered backend architecture separating domain, application, infrastructure, and API responsibilities.
- Implemented JWT authentication, password recovery, searchable and filterable course discovery, student enrollment, course reviews, protected lesson access, and instructor tools for creating courses, sections, and video lessons.
- Integrated Firebase Storage for course thumbnails and video content and developed a responsive Arabic-English interface using vue-i18n, automatic RTL/LTR handling, Vue Router, and Axios-based API communication.

PlacePulse | MERN Community Places Platform

- Built a full-stack MERN web application for discovering, reviewing, saving, and sharing real place experiences using React, Node.js, Express, MongoDB, and Mongoose.
- Implemented JWT authentication, profile editing, place/review CRUD, media uploads, collections, ownership checks, backend validation, Mongoose transactions, geocoding, and interactive maps.
- Integrated Gemini-powered review summaries, place Q&A, embeddings, and semantic search to help users find places using natural-language queries.

Toytastic | MVC E-Commerce Platform

- Developed a production-style e-commerce platform using Node.js, Express, MongoDB, Mongoose, and EJS with MVC architecture.
- Implemented session authentication, email verification, password reset, RBAC, CSRF protection, rate limiting, product CRUD, cart/checkout workflows, inventory safety, order lifecycle management, invoices, and admin moderation tools.
- Built backend workflows for stock enforcement, order status transitions, review approval, invoice generation, flash messaging, error handling, and environment-based configuration.

Technical Skills

Programming: JavaScript, TypeScript, Python, SQL, C#, C/C++,

Frontend: React, Vue.js, Next.js, HTML5, CSS3, Responsive UI

Backend: Node.js, Express.js, ASP.NET Core Web, FastAPI, REST APIs, GraphQL, WebSockets, MVC Architecture

Databases & Cloud: MongoDB, Mongoose, PostgreSQL, MS SQL Server, MySQL, Firebase

Tools: Docker, Git, GitHub, npm, Postman

Activities

- **Zewail City Science Festival 2022:** I contributed as a content creator and actor in scientific mini plays.
- **IEEE Zewail City Club:** I designed and conducted scientific experiments to illustrate key scientific concepts in different fields.